Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

III SEMESTER B.TECH. (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS, NOVEMBER-DECEMBER 2024

ELECTRICAL MACHINERY-I [ELE 2123]

REVISED CREDIT SYSTEM

Time: 3 Hours **25 November 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. A 100 kVA, 3,300 V / 200 V, 50 Hz single-phase transformer has 40 turns on the secondary. If this transformer should be used on a 25 Hz system, determine the following: (3)

a) New primary voltage when the peak flux is increased by 10 % of the rated value.

b) New kVA rating of the transformer if the rated winding currents are unaltered.

1B. Open-circuit and short-circuit tests conducted on a 4 kVA, 200 V / 400 V, 50 Hz, single-phase transformer gave the following results: (4)

OC test: 200 V, 1 A, 100 W (on LV side)

SC test: 15 V, 10 A, 85 W (on HV side)

a) Obtain the parameters of the equivalent circuit referred to the LV side.

b) Determine the voltage regulation when the transformer supplies full load at 0.8 lagging power factor.

c) Analyse the reasoning behind conduction of SC test on HV side with the LV side short-circuited.

1C. A single-phase two-winding transformer is rated for 100 kVA, 60 Hz, 12,500 V / 2,500 V. This transformer should be used as an auto-transformer. Draw any 3 possible connections for the autotransformer and determine the corresponding voltage ratios and output kVA ratings. (3)

2A. A three-phase step down transformer is connected to 6,600 V mains and draws a current of 24 A. The ratio of turns per phase is 12. Power losses may be neglected. Calculate the secondary line voltage, line current and output kVA for the following connections: (3)

a) Delta-star b) Star-delta.

2B. Sketch the torque-slip characteristic of a 3-phase squirrel cage induction motor indicating therein the starting torque, maximum torque and the operating region. Analyze and distinguish how these characteristics can be altered in a wound rotor induction motor with respect to the following aspects: (4)

a) Starting torque and its magnitude

b) Slip at which the maximum torque occurs

c) Magnitude of the maximum torque

2C. A 3-phase, 400 V, 50 Hz, 4-pole, star connected induction motor has per-phase stator impedance, $Z_1 = (0.07+j0.3)\Omega$ and per-phase rotor impedance referred to stator side, $Z_2' = (0.08+j0.3)\ \Omega$. The per phase magnetizing reactance is 10 Ω and the resistance representing core loss is 50 Ω. The slip is 4%. Using approximate equivalent circuit approach, solve for: 3

a) Stator current and power factor
b) Torque developed and
c) Gross efficiency

3A. With the help of double field revolving theory, prove that a single-phase induction motor containing only one stator winding produces no starting torque. Justify your answer with suitable characteristics. 3

3B. A 3-phase, 400 V, 3.7 kW, 7.5 A, 945 rpm, delta connected squirrel cage induction motor has per phase stator resistance of 5 Ω. This motor is subjected to no-load and blocked-rotor tests. The test results are: 4

No load test: 400 V, 3.25 A, 232 W
Blocked rotor test: 92 V, 7.5 A, 650 W

a) Determine the parameters of its approximate equivalent circuit.
b) Draw and label the approximate equivalent circuit referred to stator.

3C. A DC shunt generator on full load gives an output of 30 kW at a terminal voltage of 200 V. The armature and shunt field resistances are 0.05 Ω and 50 Ω respectively. The iron, friction and windage losses are 1,000 W. Determine the following: 3

a) Generated EMF
b) Copper losses
c) Efficiency

4A. A 6 pole long shunt DC compound generator supplies 50 A through 4 brushes at a terminal voltage of 220 V. Its armature resistance is 0.02 Ω, series field resistance is 0.04 Ω and shunt field resistance is 110 Ω. Take voltage drop per brush as 1 V. Neglect armature reaction. 3

a) Draw the connection diagram and label it fully.
b) Determine the EMF developed in the armature.

4B. A 250 V DC shunt motor on no-load runs at 1,200 RPM and draws a supply current of 5 A. The total armature and shunt field resistances are 0.2 Ω and 250 Ω respectively. When this motor draws a full-load current of 40 A, determine its full-load speed. Assume the armature reaction to weaken the flux by 3 % from no-load to full-load. 3

4C. A 250 V DC shunt motor has an armature resistance of 0.5 Ω and field resistance of 250 Ω. When driving a constant torque load at 600 RPM, the motor draws 21 A. Compute the new speed of the motor if an additional resistance of 250 Ω is inserted in its field circuit. Assume a linear magnetization characteristic. 4

5A. From fundamentals of force exerted on a conductor, derive an expression for the torque developed in a DC motor. Use this expression to justify how the motor's direction of rotation could be reversed. 3

5B. A DC series motor should never be started on no-load. Justify this statement. Explain using relevant characteristics. 3

5C. A Field's test is conducted on two mechanically coupled DC series motors. It gave the following results: 4

Motor:

Armature current = 50 A, armature voltage = 500 V, field winding voltage drop = 38 V

Generator:

Armature current = 38A, armature voltage = 400 V, field winding voltage drop = 36 V

Resistance of each armature winding is 0.2 Ω. Determine the operating efficiency of the generator for the conducted test.